

Clinical Risk Assessment Report: Patient 7

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.43x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 1.6%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.07 [Avg (45th %ile)] (-1.6%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 74.0 [Top 13%] (-1.3%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.23 [Avg (44th %ile)] (-1.0%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -0.38 [Avg (29th %ile)] (-0.9%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.84 [Avg (74th %ile)] (-0.7%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.39 [Avg (42th %ile)] (-0.6%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.13 [Avg (52th %ile)] (-0.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.4 [Avg (32th %ile)] (-0.6%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Machine Learning Feature Attribution (SHAP Decision Path)

